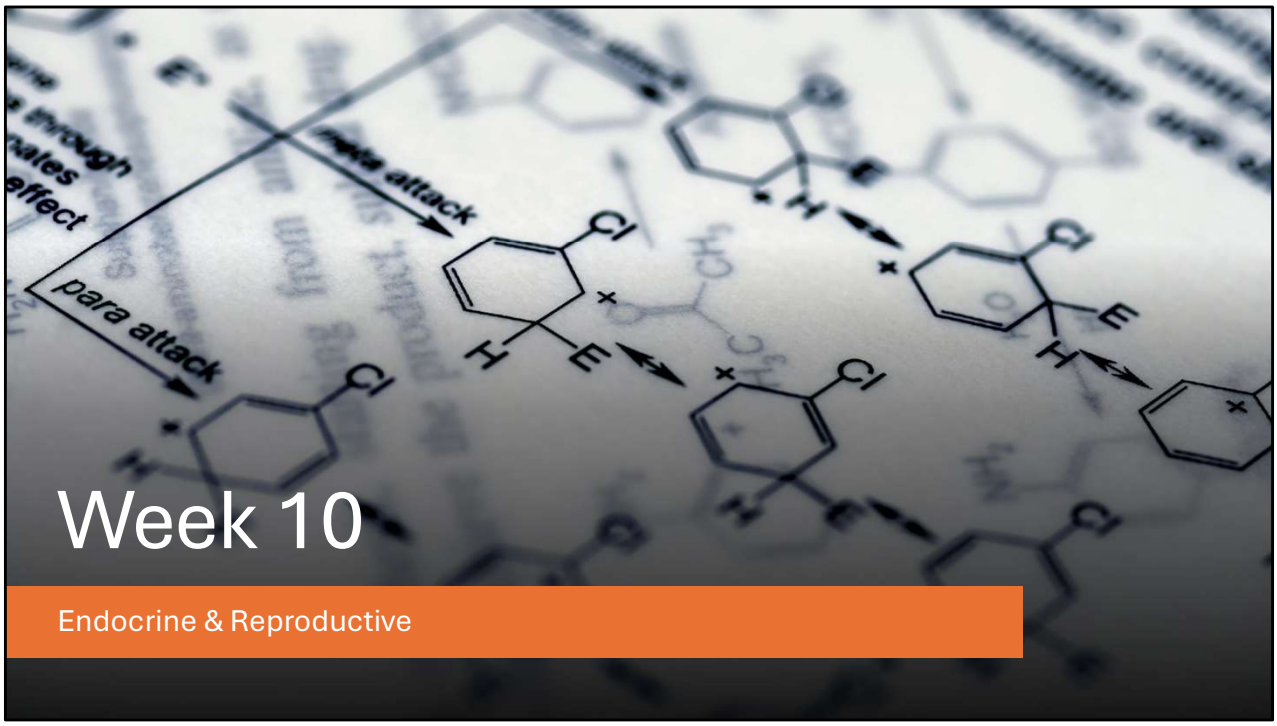


Week 10

Endocrine & Reproductive

Endocrine Review

Hormones
Pituitary Gland
Primary, Secondary, Tertiary Disorders

Which hormone is controlled by positive feedback?

A. ADH
B. PTH
C. Oxytocin
D. Cortisol

C

What causes most pituitary disorders?

A. Brain radiation
B. Benign adenomas
C. Genetics
D. Infections

B

Which of the following is not a way that hormones are regulated?

A. Neural stimulation
B. Humoral stimulation
C. Hormone stimulation
D. Muscle stimulation

D

The nurse cares for a client with acromegaly. Which sign or symptom is the nurse likely to encounter while caring for this client?

A. The client has a high-pitched, soft voice
B. The client has smooth, thin skin
C. The client's fasting glucose is 203
D. The client's BP is 95/76

C

ADH Disorders

SIADH
Diabetes Insipidus (DI)

What is the pathophysiology of diabetes insipidus?

A. Deficit of ADH
B. Excess of ADH
C. Deficit of aldosterone
D. Excess of aldosterone

A

What is a manifestation of SIADH?

A. Hypoglycemia
B. Dehydration
C. Confusion
D. Dilute urine

C

The nurse cares for a client admitted with DI. Which findings does the nurse recognize as correlating with the disorder? Select all that apply.

A. Hypertension
B. Polyuria
C. Tachycardia
D. Elevated urine specific gravity
E. Excessive thirst

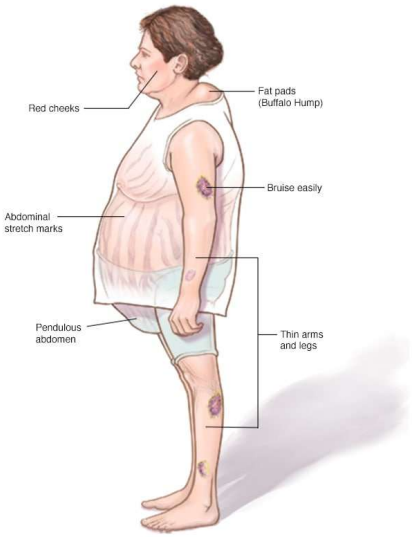
B, C, E

Adrenal Disorders

Cushing's Syndrome
Addison's Disease

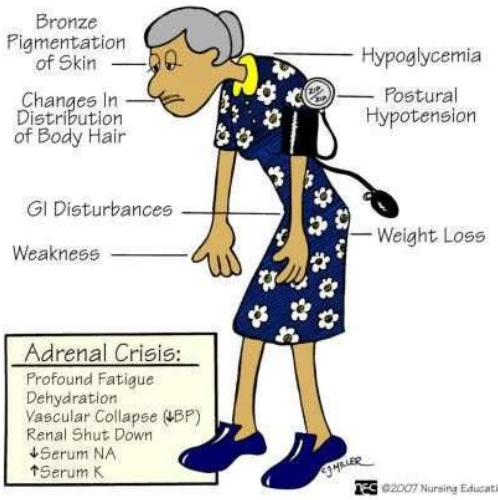
Cushing's Syndrome





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ADDISON'S DISEASE



What is a manifestation of Addison's disease?

A. Moon face
B. Buffalo hump
C. Weight loss
D. Striae on skin

B

What is the pathophysiology of Cushing's disease?

A. Deficit of ADH
B. Excessive glucocorticoids
C. Deficit of growth hormone
D. Excessive vasopressin

B

What is not a possible cause of Cushing's disease?

A. Abrupt stop of steroid use
B. Adrenal adenoma
C. Pituitary adenoma
D. Prolonged intake of steroids

A

Which of the following statements best describes an adrenal (Addisonian) crisis?

A. Adrenal crises occur when the body makes too much cortisol
B. Adrenal crises result in hyperglycemia and decreased immune response
C. Adrenal crises occur when the body is under stress and the adrenal cortex does not function
D. Adrenal crises cause dangerous hypertension

C

The nurse provides education to a client prescribed a glucocorticoid for cortisol replacement following an adrenalectomy. Which client statement requires follow up?

A. “It is normal for my weight to fluctuate 3-5 lb (1.4-2.3 kg) daily.”

B. ”My medication may require a dose change if I am stressed.”

C. “It is important for me to avoid large crowds and to physically distance.”

D. “I should use a moisturizer on my skin daily.”

A

A nurse develops a plan of care for a client who was recently diagnosed with Cushing syndrome. The nurse prioritizes reducing the risk for which potential complication of Cushing syndrome?

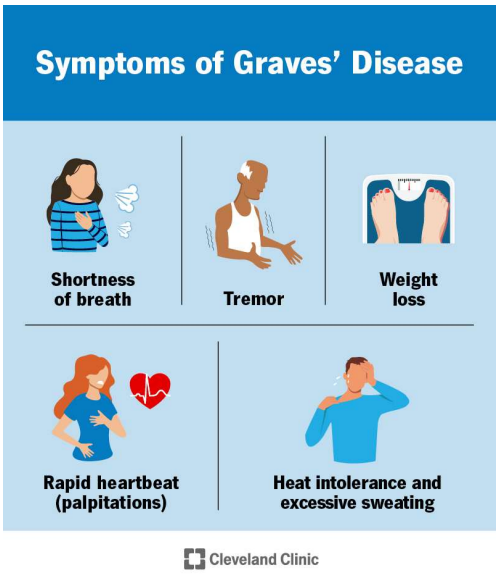
A. Risk for hypotension
B. Risk for dehydration
C. Risk for infection
D. Risk for hypoglycemia

C

Thyroid Disorders

Hyperthyroidism – Grave's Disease
Hypothyroidism – Hashimoto's Thyroiditis




Symptoms of Graves' Disease
Shortness of breath
Tremor
Weight loss
Rapid heartbeat (palpitations)
Heat intolerance and excessive sweating
Cleveland Clinic

Hypothyroidism

Low T3/T4: Hypometabolic State

Decreased BMR:
Weight gain, Dec. Appetite

Goiter (pot)



Skin, Hair, & Face

- Cold intolerance, reduced sweating
- Cool, Dry skin w/edema
- Puffy face/eyelids
- Hair loss



Cardiovasc. & Resp.

- Dec. HR & CO.
- Dec. Ventilatory responses.

NeuroMusc/Psych.

- Depression
- Slower speech/ mental funct.
- Fatigue, lethargy
- Delayed relaxation of tendon reflexes
- Proximal Myalgia



Myxedema Madness

Delusions
Hallucinations
Delirium



Myxedema Crisis

Extreme lethargy, stupor, or coma w/ hypothermia.



What is a manifestation of Grave's disease?

A. Heat intolerance
B. Weight gain
C. Non-pitting edema
D. Lethargy

A

What is a manifestation of myxedema?

A. Hyperthermia
B. Tachycardia
C. Restlessness
D. Facial puffiness

D

Diabetes Mellitus

Type I vs Type II
DKA vs HHS
Hypoglycemia

Skin Complications of DM

Dermopathy



Acanthosis nigricans



Necrobiosis lipoidica diabeticorum



What are symptoms of hypoglycemia? Select all that apply.

A. Fatigue
B. Blurred vision
C. Shakiness
D. Excessive urination
E. Glucosuria
F. Diaphoresis

A, B, C, F
A and B are more hyper but can happen with very low BS levels

A nurse is educating a patient on type II diabetes. Which statement by the patient indicates the need for further teaching?

A. “With type II, diet and exercise may help my condition.”
B. “With type II, I will inspect my feet daily.”
C. “With type II, the pancreas does not produce any insulin.”
D. “With type II, losing weight may help my condition.”

C

Which statements are true with DKA? Select all that apply.

A. Findings include ketones in the urine
B. Caused by type I diabetes
C. Findings include dehydration
D. Findings include metabolic alkalosis
E. Caused by type II diabetes
F. Findings include metabolic acidosis

A, B, C, F

The nurse cares for an older adult client with type 2 diabetes who has developed HHS. Which lab result associated with HHS is the nurse likely to observe?

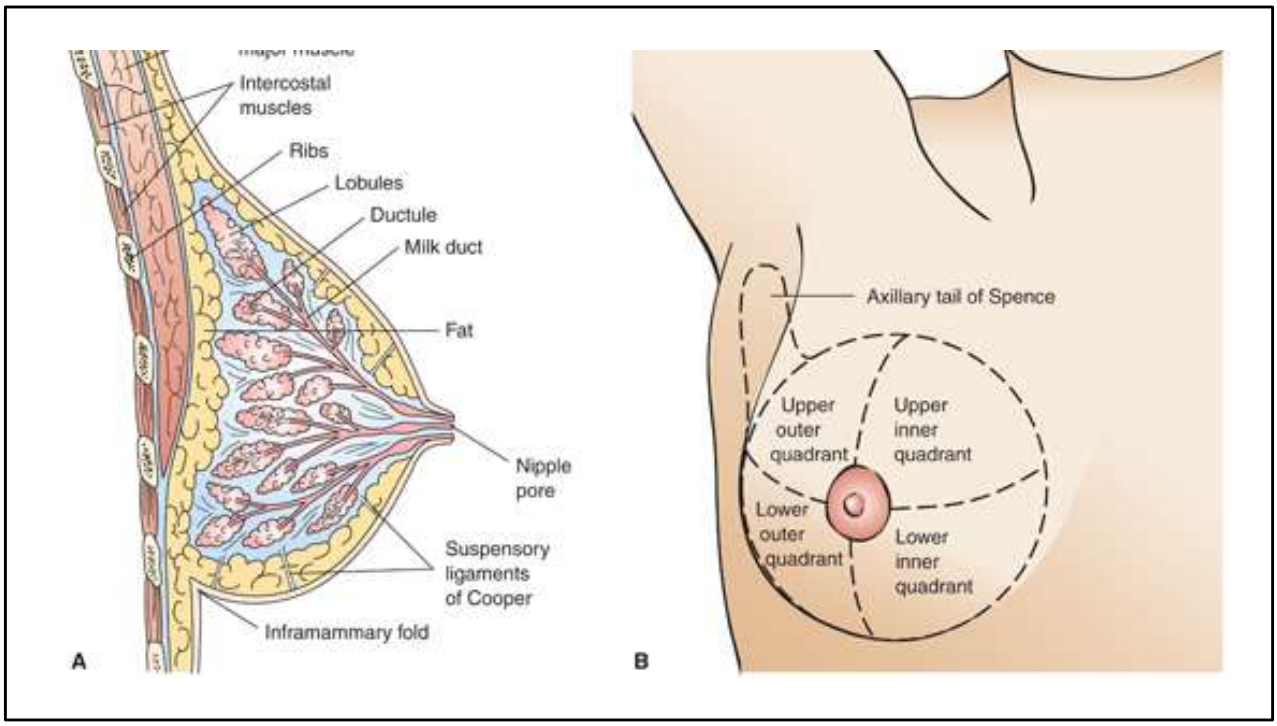
A. Serum glucose of 689
B. Significantly elevated urine ketone level
C. Serum potassium level of 2.9
D. Low serum osmolarity

A

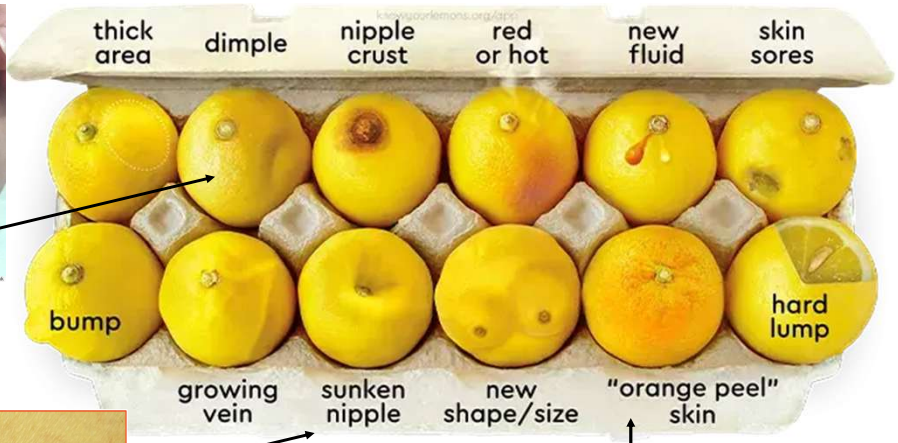

Reproductive

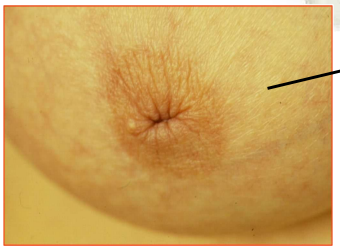
Female Reproductive System

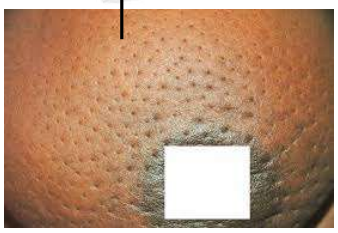
- Breast Cancer
- Menopause
- PCOS
- PID
- Uterine Prolapse
- Vaginitis
- Ovarian Cancer
- Cervical Cancer
- Uterine Fibroids
- Ovarian Cysts
- Infertility

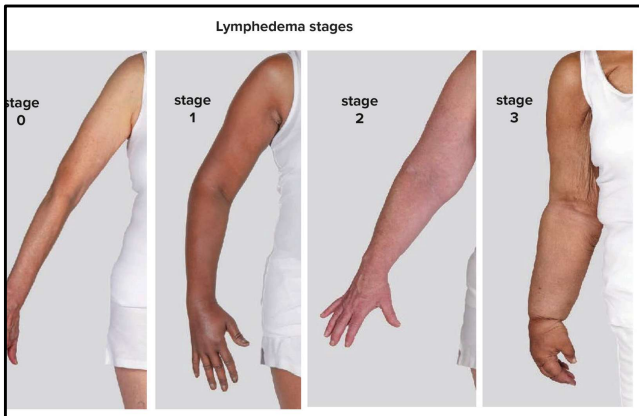

Intercostal muscles
Ribs
Lobules
Ductule
Milk duct
Fat
Nipple pore
Suspensory ligaments of Cooper
Inframammary fold
A
Axillary tail of Spence
Upper outer quadrant
Upper inner quadrant
Lower outer quadrant
Lower inner quadrant
B

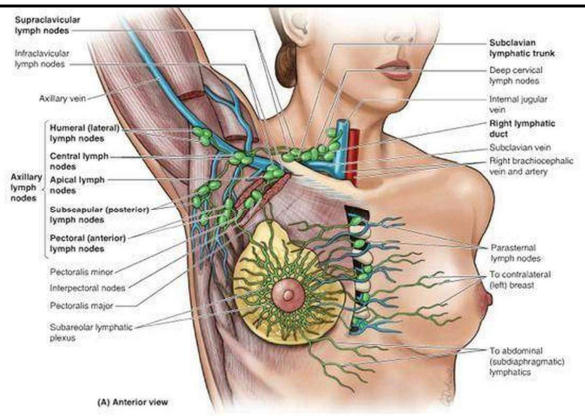



thick area
dimple
nipple crust
red or hot
new fluid
skin sores
bump
hard lump
growing vein
sunken nipple
new shape/size
"orange peel" skin






Lymphedema stages
stage 0
stage 1
stage 2
stage 3

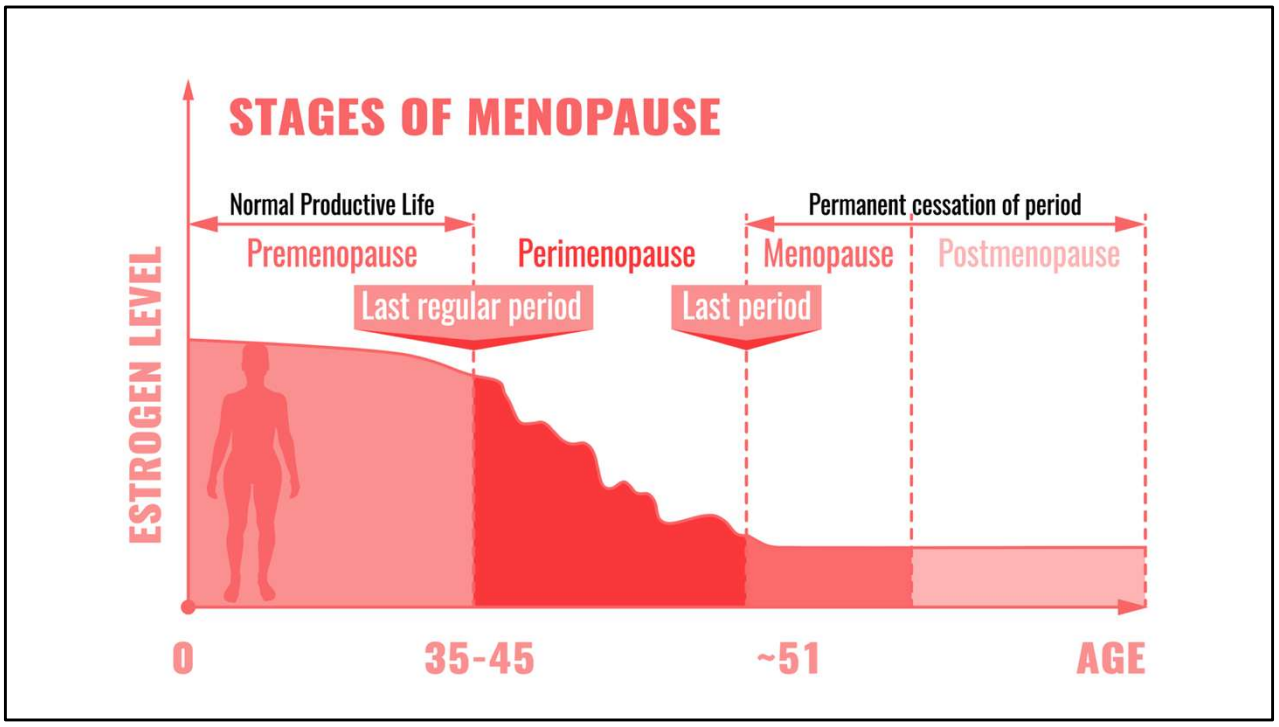

Supraclavicular lymph nodes
Infraclavicular lymph nodes
Axillary vein
Humeral (lateral) lymph nodes
Central lymph nodes
Apical lymph nodes
Axillary lymph nodes
Subscapular (posterior) lymph nodes
Pectoral (anterior) lymph nodes
Pectoralis minor
Interpectoral nodes
Pectoralis major
Subareolar lymphatic plexus
Subclavian lymphatic trunk
Deep cervical lymph nodes
Internal jugular vein
Right lymphatic duct
Subclavian vein
Right brachiocephalic vein and artery
Parasternal lymph nodes
To contralateral (left) breast
To abdominal (subdiaphragmatic) lymphatics
(A) Anterior view

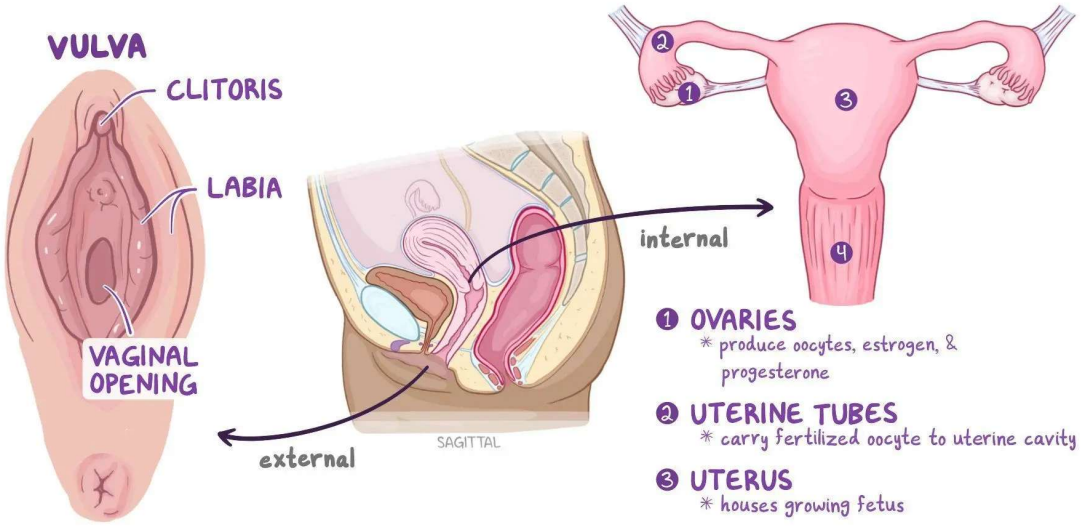
Lymphedema

The nurse discusses the screening recommendations for breast cancer. What information does the nurse include?

A. Mammography for women should start around the age of 40
B. Women with a family history of breast cancer should begin mammograms at age 40
C. Clinical breast examination is best for early cancer detection
D. Women with no family history under age 50 are at low risk, so no screening is needed

A


STAGES OF MENOPAUSE
Normal Productive Life
Permanent cessation of period
Premenopause
Perimenopause
Menopause
Postmenopause
Last regular period
Last period
ESTROGEN LEVEL
0
35-45
~51
AGE



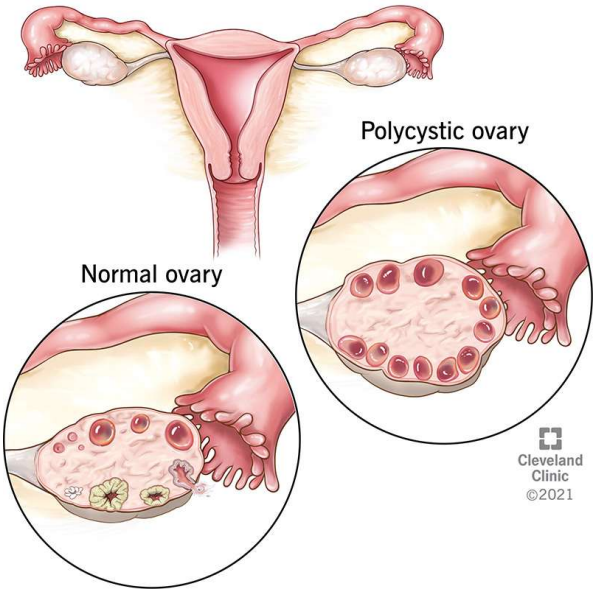
❶ **OVARIES**
* produce oocytes, estrogen, & progesterone

❷ **UTERINE TUBES**
* carry fertilized oocyte to uterine cavity

❸ **UTERUS**
* houses growing fetus

❹ **VAGINA**
* involved in intercourse & fetal delivery

PCOS



The nurse cares for a client diagnosed with polycystic ovary syndrome (PCOS). What signs or symptoms is the nurse likely to assess?

A. The client has no eyebrow or eyelash hair
B. The client has low serum testosterone
C. The client reports light periods every two weeks
D. The client has acne of the face and chest

D

FACTORS PREDISPOSING WOMEN TO DEVELOPMENT OF PID

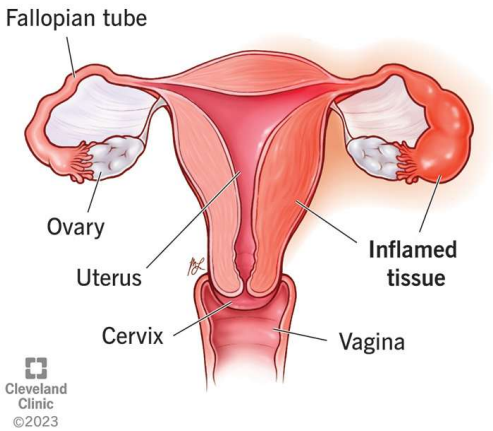
- An age of 16 to 24 years
- Unmarried status
- History of multiple sexual partners
- Previous history of PID

- **Complications**
- Abortion
- Infertility

PELVIC INFLAMMATORY DISEASE (PID)

- An inflammation of the upper reproductive tract that involves
  - The uterus (endometritis)
  - The Fallopian tubes (salpingitis)
  - The ovaries (oophoritis)
- A polymicrobial infection; cause varies by geographic location and population

Pelvic inflammatory disease (PID)

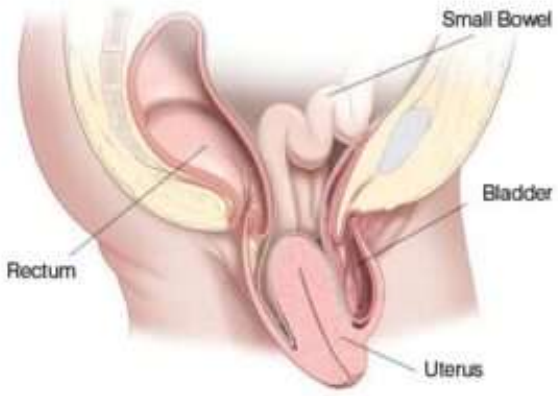


The nurse reflects on a patient who presented to the clinic with complaints of irregular menstrual cycles and pelvic discomfort. She reviews the reported symptoms and laboratory findings. Select to indicate if the findings indicate the potential for PCOS, hypothyroidism, or PID. Each finding may indicate more than one condition.

Assessment Finding	PCOS	PID	Hypothyroidism
Irregular menses			
Fatigue			
Weight gain			
Positive chlamydia			
Hirsutism			
Pelvic tenderness			
Dry skin and brittle hair			
Elevated TSH with low free T4			

1. All
2. Hypothyroid
3. PCOS, hypothyroid
4. PID
5. PCOS
6. PID
7. Hypothyroid
8. Hypothyroid

Uterine Prolapse



- **Vaginitis**
  - Causes: Candida, bacterial vaginosis
  - Symptoms: Itching, discharge, irritation

The nurse educates a group of postmenopausal women about gynecological health. The nurse identifies which warning sign of gynecological cancer?

A. Pain during intercourse
B. Vaginal discharge that is clear
C. Pain when urinating
D. Vaginal bleeding

D

- **Ovarian Cysts**
  - Types: Functional cysts, dermoid cysts
  - Symptoms: Pelvic pain, bloating
  - Management: Observation, surgical removal if symptomatic

- **Uterine Fibroids**
  - Benign tumors of the uterine muscle
  - Symptoms: Heavy bleeding, pelvic pain, infertility
  - Treatment: Hormone therapy, surgical removal

CAUSES OF MALE INFERTILITY

- Varicocele
- Ejaculatory dysfunction
- Hyperprolactinemia
- Hypogonadotropic hypogonadism
- Infection
- Immunologic problems (*i.e.*, antisperm antibodies)
- Obstruction
- Congenital anomalies

FEMALE FACTORS AFFECTING FERTILITY #1

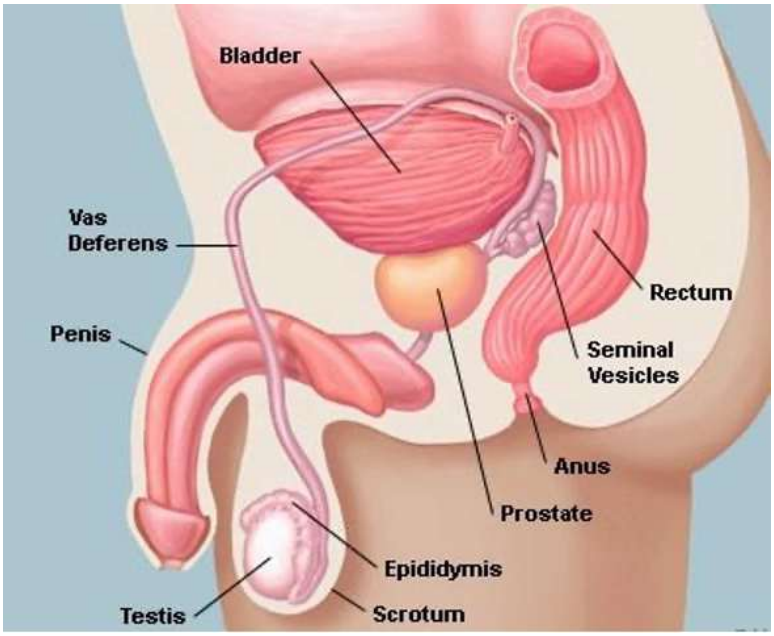
- Must be able to
  - Production and release of a mature ovum capable of being fertilized
  - Production of cervical mucus that assists in sperm transport and maintains sperm viability in the female reproductive tract
  - Patent fallopian tubes with the motility potential to pick up and transfer the ovum to the uterine cavity

FEMALE FACTORS AFFECTING FERTILITY #2

- Development of an endometrium that is suitable for the implantation and nourishment of a fertilized ovum
- A uterine cavity that allows for growth and development of a fetus

Male Reproductive System

- Hypospadias
- Urethritis
- Prostatitis
- BPH
- Phimosis + Paraphimosis
- Erectile Dysfunction
- Prostate Cancer
- Orchitis
- Varicocele
- Testicular Torsion
- Testicular Cancer

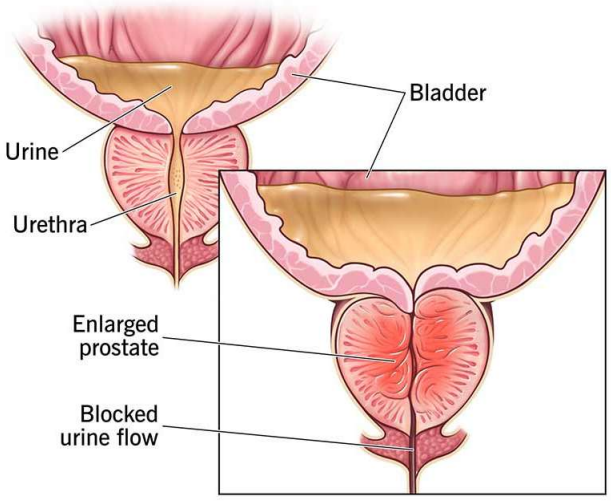

Bladder
Vas Deferens
Penis
Rectum
Seminal Vesicles
Anus
Prostate
Epididymis
Testis
Scrotum



- **Hypospadias**
  - Definition: Abnormal urethral opening along the underside of the penis
  - Causes: Genetic factors, hormone imbalances during fetal development
  - Clinical Presentation: Curved penis, difficulty urinating
  - Management: Surgical correction

URETHRITIS & PROSTATITIS

Benign Prostatic Hyperplasia



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The nurse in a primary care clinic oversees a student giving prostate screening education to a client. The nurse intervenes when the student makes which statement?

A. “Screening is continued as long as life expectancy is at least 10 years.”
B. “A false positive may occur with the prostate-specific antigen test.”
C. “Having a family history of prostate cancer can increase your risk.”
D. “Routine prostate screening is recommended starting at the age of 40.”

D

A nurse develops a care plan for a male client. The nurse includes which statements about prostate cancer? Select all that apply

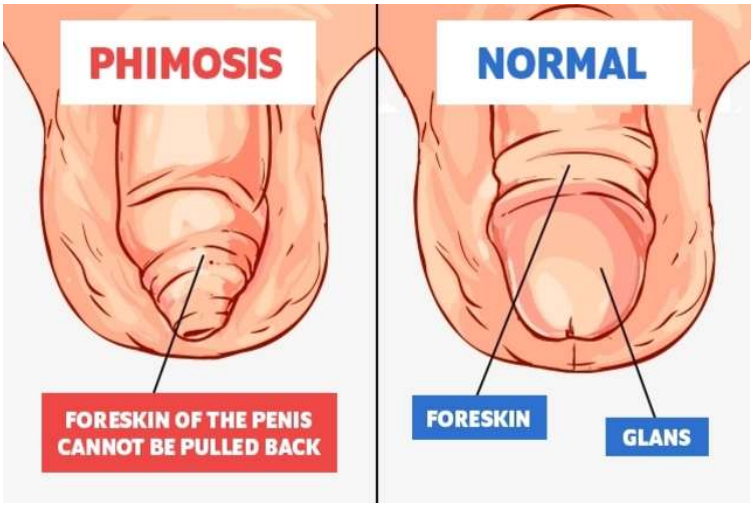
A. Prostate cancer may be indicated through a test for specific antigens
B. Having a relative with prostate cancer is a risk factor
C. Symptoms may mimic those of benign prostatic hypertrophy
D. Prostate cancer may be detected by digital rectal exam
E. Prostate cancer is usually an adenocarcinoma

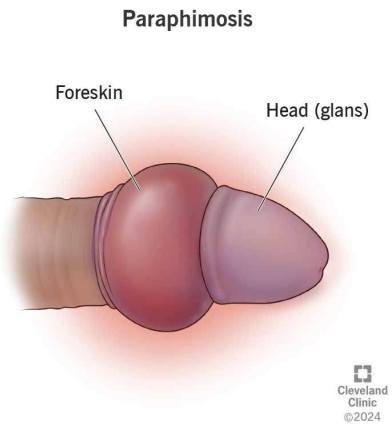
All

A nurse develops a health information session targeting clients at higher risk for prostate cancer. What clients does the nurse teach? Select all that apply.

A. Transgender females on hormone therapy
B. Clients who are prescribed finasteride
C. Clients taking folic acid supplements
D. Clients who are African American
E. Clients who are over 50 years of age

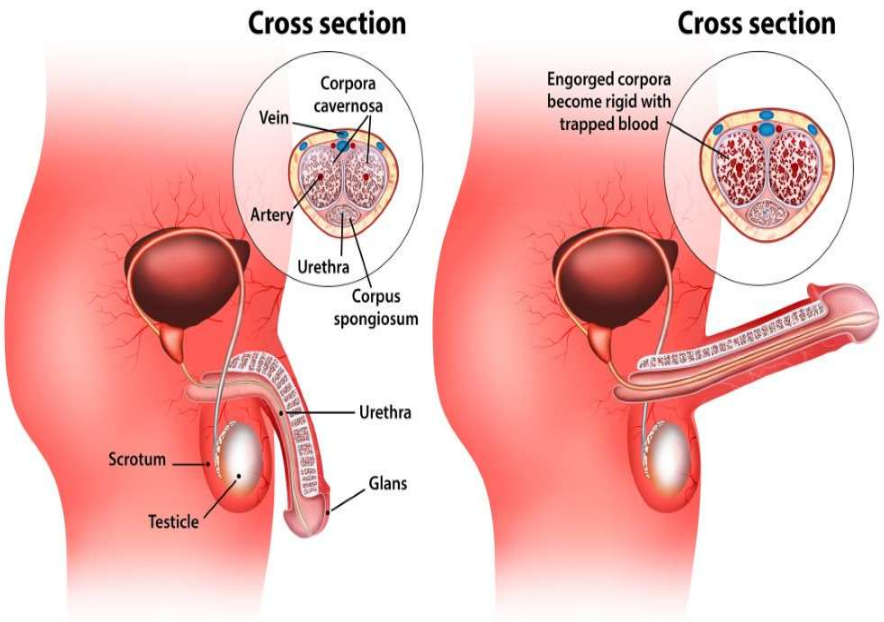
C, D, E





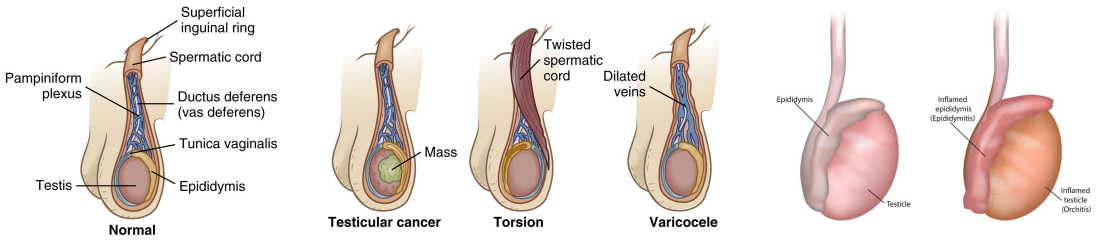
All the Phimosisises

Erectile Dysfunction



PROSTATE CANCER

Testicle Things



The nurse interviews a client suspected of having testicular cancer. The nurse knows that which factors increase the client's risk of testicular cancer? Select all that apply.

A. History of undescended testicle
B. Eight sexual partners last year
C. Age of 19 years old
D. Brother had testicular cancer
E. Hispanic and Asian ethnicity

A, C, D

A 45-year-old male with unilateral testicular cancer is scheduled for an orchiectomy. Which statements indicate understanding of the education provided by the nurse for the procedure? Select all that apply.

A. “I will wear a supportive device for the first 2-3 days after the procedure.”
B. “I will need to take hormones for the rest of my life after the procedure.”
C. “I will need to have special counseling sessions after recovery from the procedure.”
D. “I will not be able to have children in the future once this surgery is complete.”
E. “The procedure is often done in an outpatient setting.”

A, E